

PNETLab v8 Master Architecture & Deployment Guide

Complete Technical Blueprint: Network Persistence, 100% Standalone Isolation, Appliance Packaging & OS Stability

Target Platform: Ubuntu 26.04 LTS (Resolute) Verified VM: R9-PNET (192.168.1.23)

Release Stream: 1-P-872-ME-STABLE (v6.8.72) Generated: September 2026

SECTION 1 Executive Summary of Resolved Issues

This master guide consolidates all architectural solutions, troubleshooting steps, and design recommendations developed for PNetLab v8 running on modern Linux kernels. All fixes have been validated on host `192.168.1.23` and committed directly to the core standalone bundle.

Network Persistence & IP Locking

Eliminated interface dropping and bridge creation failures on reboot. Cloud-init overrides disabled; native `systemd-networkd` + early-boot enforcer configured.

VERIFIED PERSISTENT

Autonomous Standalone Bundle

Zero external dependency on Codeberg. 38 local Debian packages (6.8.72) and all 16 database tables embedded for air-gapped environments.

100% OFFLINE

Workbench Routing & Schema Fixes

Resolved `lab_session_lid` and missing `wiresharks` table exceptions. Apache `/legacy/topology` rewrite enabled.

E2E GUI FUNCTIONAL

Multi-Layer Update Freeze

Protected core binaries via `Pin-Priority:-1`, `apt-mark hold`, DNS blackhole, and immutable safety barriers.

SAFE FOR APT UPGRADE

SECTION 2 Static IP & Network Boot Architecture

Modern Ubuntu releases utilize **Netplan** and `systemd-networkd` rather than legacy `ifupdown`. The table below illustrates the 5-layer network persistence architecture implemented to guarantee that static IPs and bridge devices (`pnet0`) never drop across reboots:

Layer	Configuration File / Service	Technical Purpose
Layer 1: Cloud-Init Shield	<code>/etc/cloud/cloud.cfg.d/99-disable-network-config.cfg</code>	Permanently disables cloud-init from regenerating network configs and resetting IP addresses on boot.
Layer 2: Kernel Module Preload	<code>/etc/modules-load.d/pnetlab.conf</code>	Preloads <code>bridge</code> , <code>8021q</code> , <code>stp</code> , and <code>tun</code> kernel modules at early init, avoiding netdev creation errors.
Layer 3: Native Networkd Policy	<code>/etc/systemd/network/10-eth0-static.network</code>	Instructs <code>systemd-networkd</code> to explicitly bind <code>192.168.1.23/24</code> with <code>DHCP=no</code> , stopping lease-timeout flushes.
Layer 4: Netplan Authority	<code>/etc/netplan/01-pnetlab-netcfg.yaml</code>	Consolidates configuration into a single authoritative YAML file; backs up conflicting installer/cloud configs.
Layer 5: Early-Boot Enforcer	<code>pnetlab-boot-network.service</code>	Runs before <code>network-online.target</code> to assert physical link UP, enable promiscuous mode, and verify default routes.

Native Systemd-Networkd Static Stanza (Production Example)

```
[Match]
Name=eth0 ens33 enp*

[Network]
Address=192.168.1.23/24
Gateway=192.168.1.1
DNS=8.8.8.8 1.1.1.1
DHCP=no
IPv6AcceptRA=no

[Route]
Destination=0.0.0.0/0
Gateway=192.168.1.1
Metric=100
```

SECTION 3 100% Standalone Offline Deployment Model

To eliminate reliance on external servers (such as Codeberg), the `1-P-872-ME-STABLE` repository contains all assets required for a fully autonomous, offline install:

Key Offline Assets Included:

- **Binary Package Pool** (`debian/pool/resolute/main/`): 38 production `.deb` packages covering PNetLab v8 (6.8.62 – 6.8.72), Guacd, QEMU, VPCS, Docker, and Bridge DKMS.
- **Complete Database Schemas** (`schema/`): Shipped SQL definitions for `pnetlab_db` and Apache Guacamole (`guacdb`).
- **Offline Mode Database Seeding**: Automatically sets `ctrl_offline_mode = 1` and `ctrl_default_mode = 'offline'` in MySQL so the web UI never attempts external license or store calls.
- **Self-Contained Installer** (`install.sh`): Installs directly from local disk with zero external network fallbacks.

How to Deploy on a Clean / Air-Gapped Machine:

```
# 1. Package the standalone bundle from the host:
tar -czf pnetlab-standalone-v8.tar.gz 1-P-872-ME-STABLE/

# 2. Extract on target Ubuntu 26.04 VM:
sudo mkdir -p /opt/pnetlab
sudo tar -xzf pnetlab-standalone-v8.tar.gz -C /opt/pnetlab/

# 3. Run the offline installer:
cd /opt/pnetlab/1-P-872-ME-STABLE
sudo bash install.sh
```

SECTION 4 Appliance Packaging Strategy: OVA vs. ISO vs. QCOW2

To avoid repeating OS installation, partitioning, and package compilation across different lab environments, the following distribution methods are recommended:

Deployment Method	Deployment Time	Effort Level	Best Suited For	Architectural Assessment
1. Pre-built OVA Appliance	< 60 seconds	Zero Setup	VMware Workstation, Fusion, ESXi	Top Recommendation. 1-click import; hardware virtualization (VT-x) and disk fixes pre-baked.
2. Proxmox QCOW2 Image	< 60 seconds	Low	Proxmox VE / KVM Clusters	High-performance raw disk image; directly importable via <code>qm importdisk</code> .
3. Standalone Offline Tarball	3 – 5 minutes	Low	Existing Linux installations	Installs complete PNetLab stack on any already-running Ubuntu 26.04 instance.
4. Custom Automated ISO	5 – 10 minutes	Medium	Bare-Metal Physical Servers	Automates Ubuntu OS install from bootable media; best for dedicated physical servers.

Exporting the Current VM to OVA (PowerShell Command)

Because VMware OVF Tool is installed on your host, you can export your verified golden VM directly:

```
& "C:\Program Files\VMware\VMware Workstation\OVFTool\ovftool.exe" ^
"C:\Users\azamb\Documents\Virtual Machines\R9-PNET\R9-PNET.vmx" ^
"C:\Users\azamb\Desktop\PNetLab-v8-Ubuntu26-Golden-Master.ova"
```

SECTION 5 Update Freeze & Safe OS Maintenance Policy

Network emulator platforms (PNetLab, EVE-NG, Cisco CML) interface directly with kernel networking and DKMS drivers. Uncontrolled OS package upgrades can break custom kernel ABIs, PHP APIs, or QEMU socket formats.

Multi-Layer Update Freeze Mechanisms Active:

- **APT Pin-Priority -1** (`/etc/apt/preferences.d/99-pnetlab-block-updates.pref`): Assigns negative priority to all PNetLab packages, ensuring APT never selects them as upgrade candidates.
- **DPKG Status Hold**: All 8 PNetLab core packages (`pnetlab` , `pnetlab-qemu` , `pnetlab-guacd` , `pnetlab-vpcs` , `pnetlab-schema` , `pnetlab-bridge-dkms` , etc.) are locked on `apt-mark hold` .
- **CLI Updater Lockdown** (`/usr/bin/pnetlab-update`): Replaced with an immutable (`chattr +i`) security exit barrier.
- **Systemd Masking**: Background update timers and image watchers are masked.
- **DNS Blackhole**: Telemetry and store domains redirected to `127.0.0.1` in `/etc/hosts` .

Is `sudo apt upgrade` Safe on the VM?

YES. Running `sudo apt update && sudo apt upgrade -y` is completely safe. Generic Ubuntu security patches (e.g. `apparmor` , `openssl` , `grub`) will update cleanly, while all PNetLab components remain strictly protected and unchanged.

SECTION 6 Clean Boot Architecture & Troubleshooting

The table below details the services that caused red `[FAILED]` errors during boot and the permanent resolutions applied:

Service / Component	Root Cause on VM Boot	Permanent Resolution
<code>multipathd.service</code>	Enterprise SAN/Fibre-Channel disk controller looking for multi-path storage on single virtual SCSI disks.	Masked <code>multipathd.service</code> and <code>multipathd.socket</code> .
<code>keyboard-setup.service</code>	Legacy script <code>/lib/console-setup/keyboard-setup.sh</code> missing in modern systemd.	Installed <code>keyboard-configuration</code> and masked obsolete service.
<code>systemd-networkd-wait-online</code>	Stalled boot timeout waiting for unattached virtual interfaces.	Masked wait-online service for instantaneous boot.
<code>udhcpd.service</code>	Attempted global boot start instead of on-demand execution inside lab topologies.	Disabled standalone daemon; managed dynamically by PNetLab wrappers.

Verified Clean Boot Status

```
root@r9-pnet:~# systemctl --failed
UNIT LOAD ACTIVE SUB DESCRIPTION

0 loaded units listed.
```